

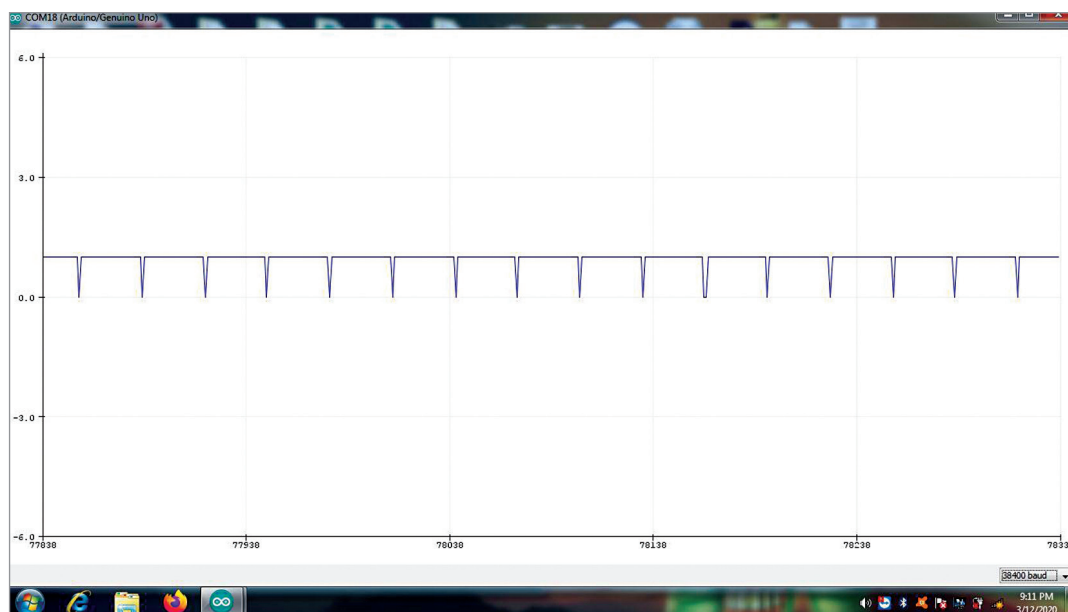


Fig. 3: Sensor pulse reading on the Arduino plotter

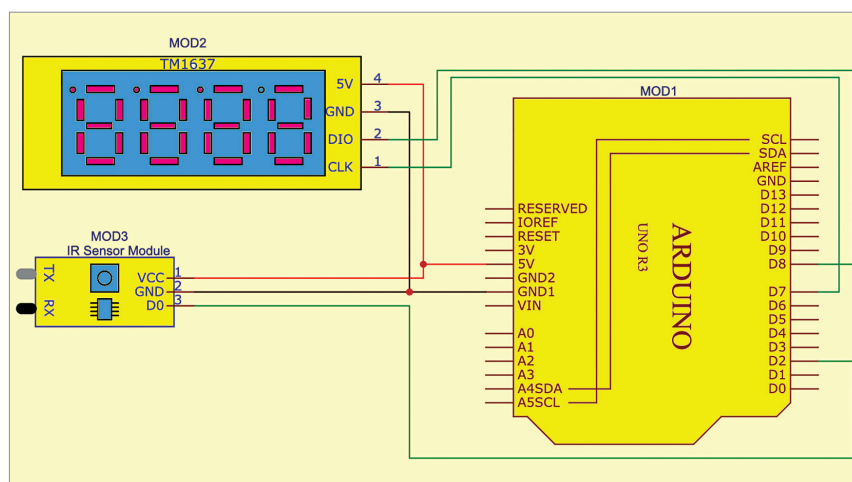


Fig. 4: Circuit for tachometer

When a 5V supply is applied to the module, the IR LED emits infrared rays. The power status is indicated by a miniature green LED on the PCB.

The photodetector receives reflected infrared rays. If a reflecting surface is present at a distance of about 2.5cm to 5cm, the reflected signal is amplified, which lights another on-board miniature green LED. Thus, one mini green LED indicates power on, while the other indicates reflection from a target.

As shown in the author's prototype (see the right side of Fig. 1), when the IR module is placed close to the fan blades without touching

them, the rotating blades reflect infrared light back to the sensor. This produces a series of pulses at the output pin as the fan runs. These pulses are illustrated in Fig. 3.

Circuit diagram

Fig. 4 shows the circuit diagram of the tachometer device. It is built around the Arduino Uno (MOD1), TM1637-based LED segment display (MOD2), and IR sensor module (MOD3), along with a few other components.

The connections to the IR module are Vcc, ground, and the pulse output pin D0. The Vcc

and ground are supplied from the Arduino board.

To provide the 5V supply to both modules, a small breadboard is used to distribute the 5V line.

The pulse output pin from the module (D0) is connected to Arduino pin D2. The CLK and DIO pins of the TM1637 display are connected to Arduino Uno pins D8 and D9.

Software

The program reads the output of an IR sensor connected to an Arduino and displays its status on the serial monitor.

In the loop() function, the Arduino continuously reads the sensor output and prints the value (high or low) to the serial monitor. Generally, high indicates that no object is detected, while low indicates detection due to reflected infrared light.

This program is useful for monitoring IR sensor pulses in applications such as object detection or tachometer measurements.

This pulse output can be observed on the Arduino serial plotter after uploading the main program. Fig. 5 shows the code snippet of the main program.

After uploading the main program, open the serial plotter from the 'Tools' menu and set the baud rate shown at the bottom-right corner to 9600.

As the fan blades rotate, the reflected infrared signals generate pulses that can be counted. By using a timer, the total number of pulses can be counted over a fixed time and displayed as pulses per minute.

For a small fan with four blades, four pulses are produced for each